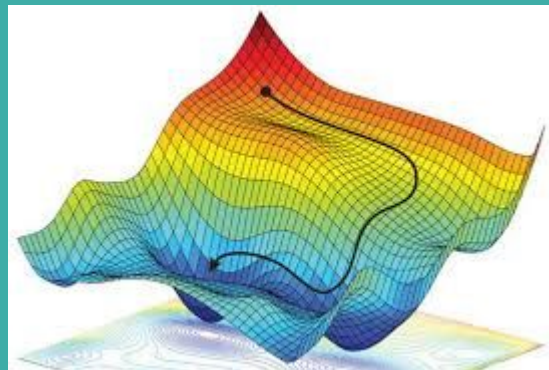


Deep Learning

CSE641/ECE555

Lecture 16 | Take your own notes during lectures

Vinayak Abrol <abrol@iiitd.ac.in>



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY **DELHI**

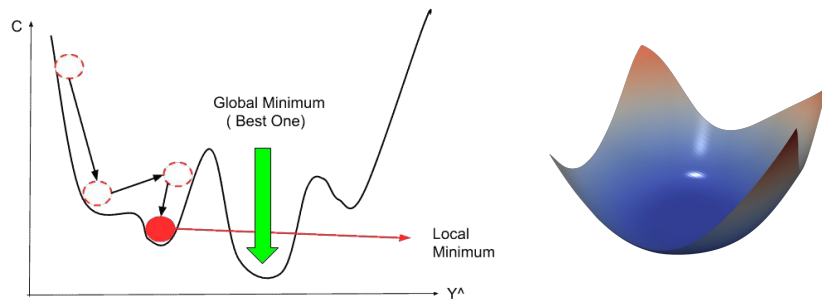
Loss Landscape Visualizations

DNN Loss function and trainable parameters

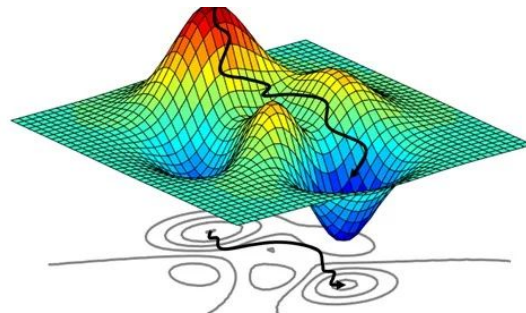
High dimensional loss function

$$\theta := \{W^{(\ell)}, b^{(\ell)}\}_{\ell=1}^L \quad |\theta| \sim n^2 L$$

$$\mathcal{L}(\theta; X, Y) = (2m)^{-1} \sum_{j=1}^m \sum_{i=1}^{n_L} (H(x_{i,j}; \theta) - y_{i,j})^2$$



The shape of $\mathcal{L}(\theta)$
and our knowledge about a good initial
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strongly influence our ability to learn the
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Loss Landscape Visualizations

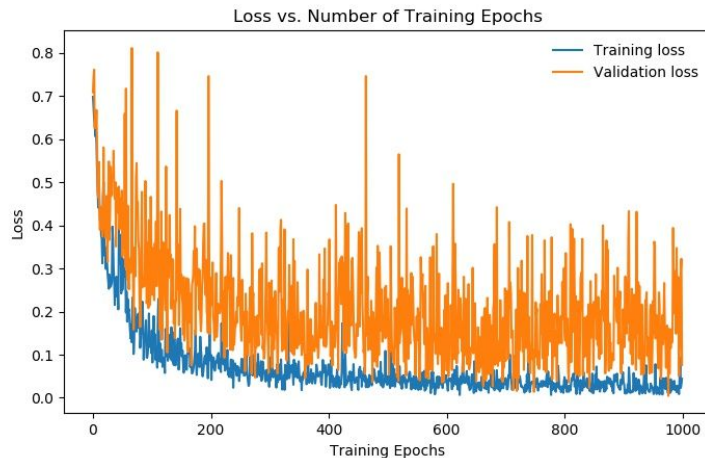
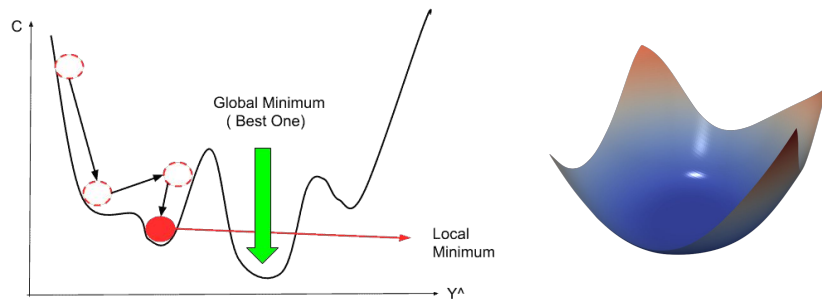
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Landscape loss function: VGG9

One dimensional views of a loss landscape

DNN loss $\mathcal{L}(\theta)$
between two minimizers,
 $\theta^s(1 - \alpha) + \alpha\theta^l$
trained with small and large batches;

Horizontal axis α
in (a) and (d).

<http://papers.nips.cc/paper/7875-visualizing-the-loss-landscape-of-neural-nets.pdf>

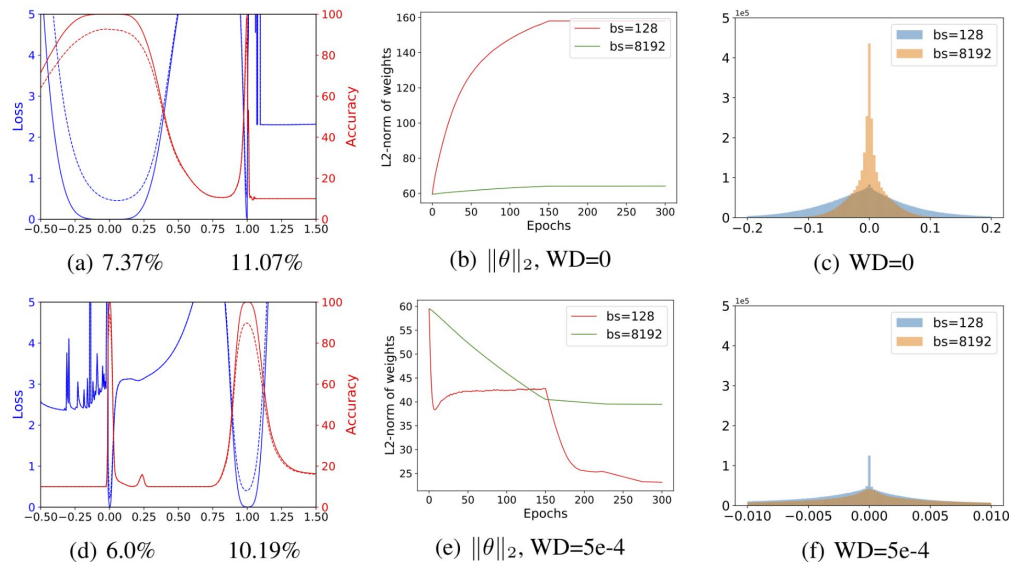


Figure 2: (a) and (d) are the 1D linear interpolation of VGG-9 solutions obtained by small-batch and large-batch training methods. The blue lines are loss values and the red lines are accuracies. The solid lines are training curves and the dashed lines are for testing. Small batch is at abscissa 0, and large batch is at abscissa 1. The corresponding test errors are shown below. (b) and (e) shows the change of weights norm $\|\theta\|_2$ during training. When weight decay is disabled, the weight norm grows steadily during training without constraints (c) and (f) are the weight histograms, which verify that small-batch methods produce more large weights with zero weight decay and more small weights with non-zero weight decay.

Landscape loss function: VGG9

One & Two dimensional views

Impact of training rate weight decay and batch size on level curves of $\mathcal{L}(\theta^* + \alpha\delta + \beta\eta)$

Larger batch size narrows the loss function.

Weight decay broadens the loss function.

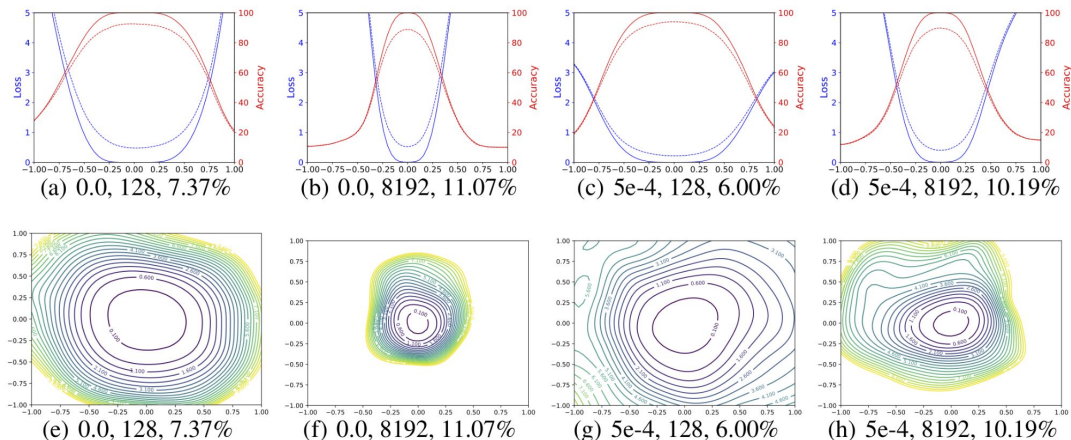
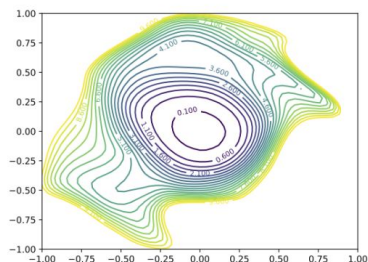


Figure 3: The 1D and 2D visualization of solutions obtained using SGD with different weight decay and batch size. The title of each subfigure contains the weight decay, batch size, and test error.

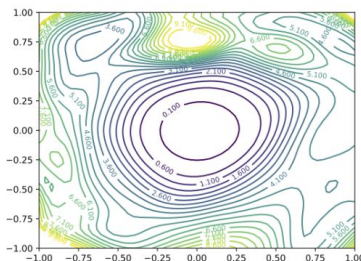
<https://arxiv.org/pdf/1412.6544>

<https://arxiv.org/pdf/1612.04010>

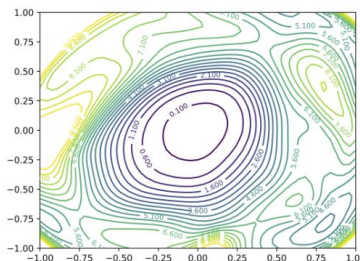
Loss landscape example: ResNet



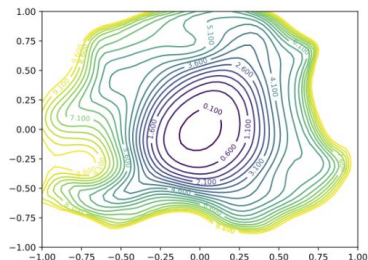
(a) ResNet-20, 7.37%



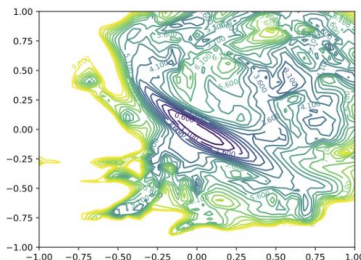
(b) ResNet-56, 5.89%



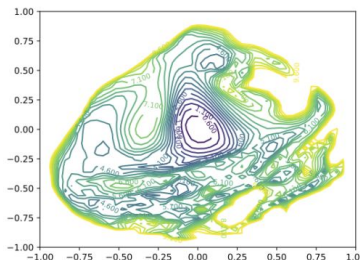
(c) ResNet-110, 5.79%



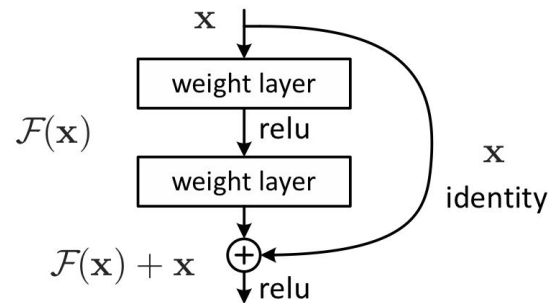
(d) ResNet-20-NS, 8.18%



(e) ResNet-56-NS, 13.31%



(f) ResNet-110-NS, 16.44%



2D visualization of the loss surface of ResNet and ResNet-noshort with different depth.

Landscape loss function: VGG9

3D dimensional views

Reconsider $\mathcal{L}(\theta^* + \alpha\delta + \beta\eta)$

Directions are sampled from a random Gaussian distribution.

While the “random directions” approach to visualization is simple, it fails to capture the intrinsic geometry of loss surfaces.

Reason: Scale Invariance such as due to weights in ReLU Nets, BN or other choice of building blocks.

Sharp Minima Can Generalize For Deep Nets

This scale invariance can be used to build pairs of equivalent networks (small and large) that have different apparent sharpness.

<https://arxiv.org/pdf/1703.04933>

Landscape loss function: VGG9

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$$\theta(\lambda) = \theta_1 + \lambda \frac{\theta_0 - \theta_1}{\|\theta_0 - \theta_1\|}$$

$$d_{i,j} \leftarrow \frac{d_{i,j}}{\|d_{i,j}\|} \|\theta_{i,j}\|$$

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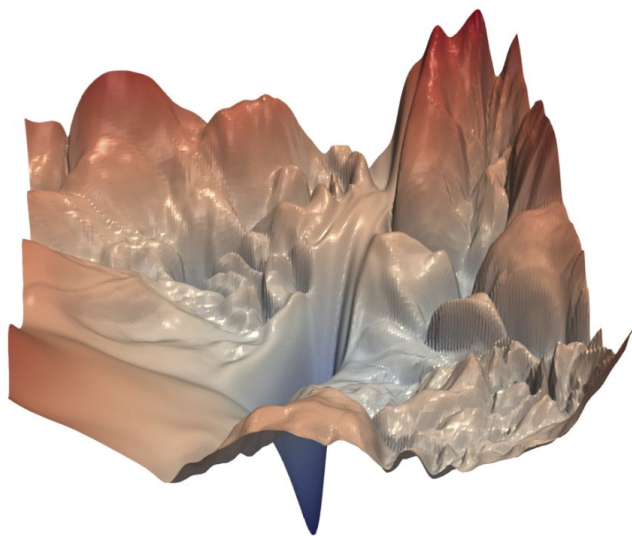
Sharp Minima Ca
This scale invaria
<https://arxiv.org/p>

Authors claim that filter-wise normalization works the best as compared to layer-wise normalization or no normalization, and show that filter normalization produces superior correlation between sharpness and generalization error.

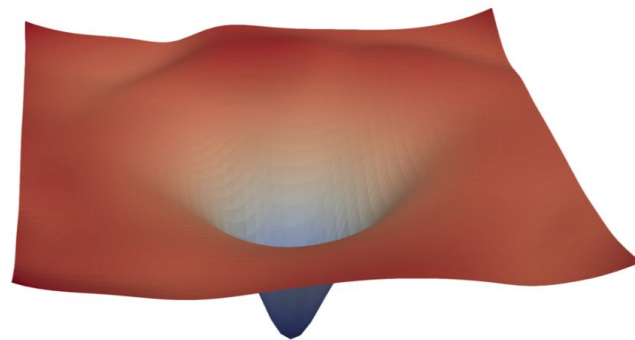
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Loss landscape example: ResNet52



(a) without skip connections

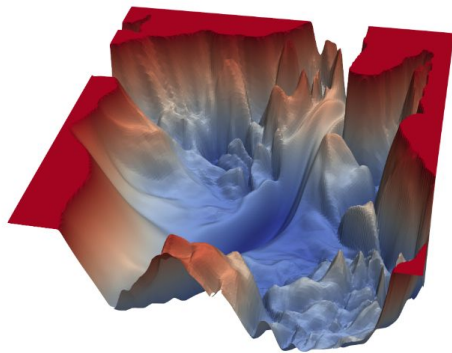


(b) with skip connections

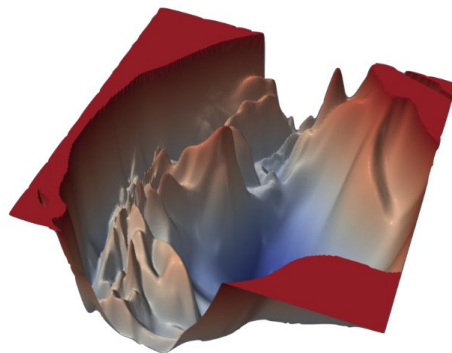
Figure 1: The loss surfaces of ResNet-56 with/without skip connections. The proposed filter normalization scheme is used to enable comparisons of sharpness/flatness between the two figures.

Loss landscape examples

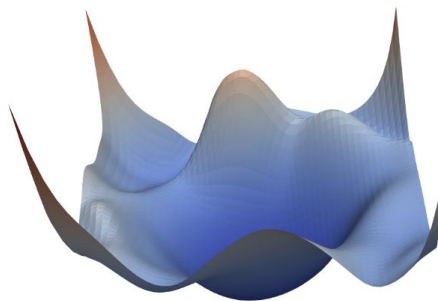
VGG-56



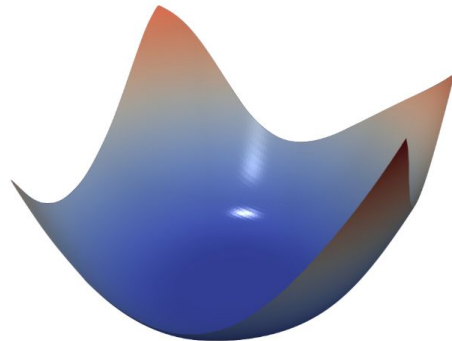
VGG-110



Renset-56



Densenet-121



Loss landscape: Impact of Width

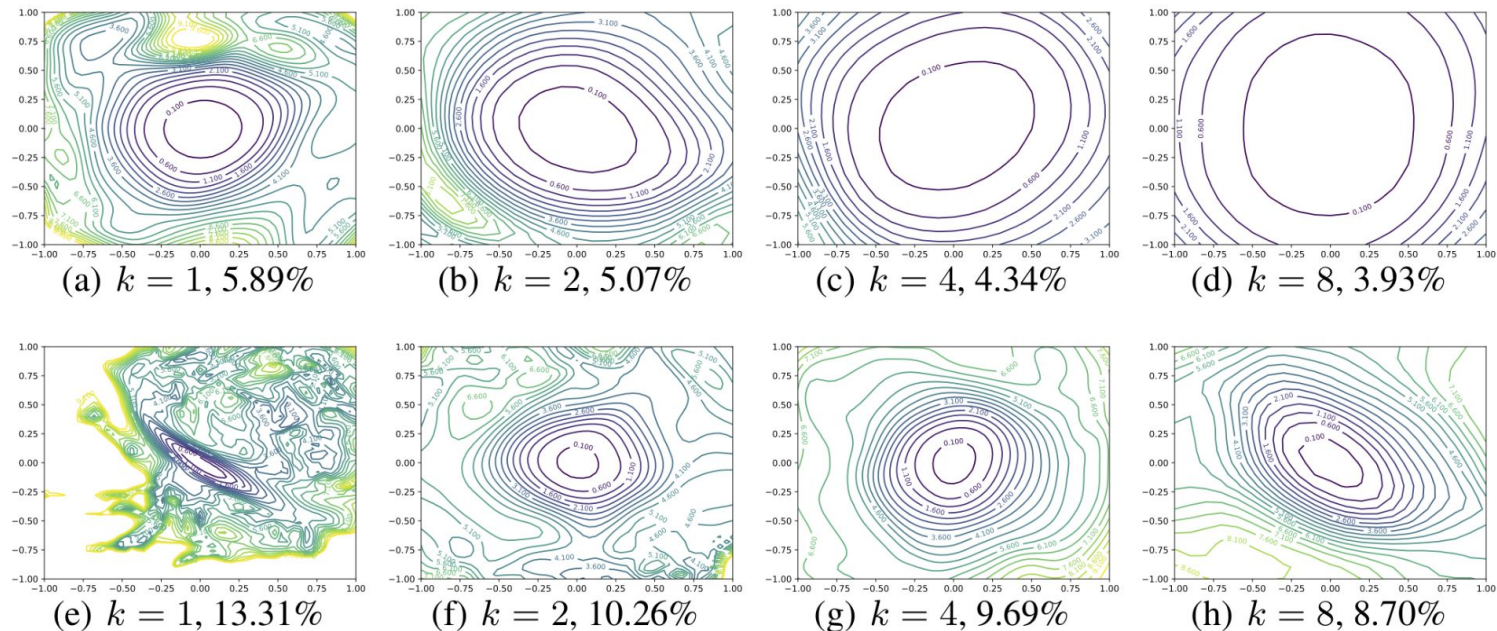
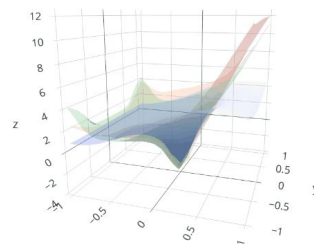
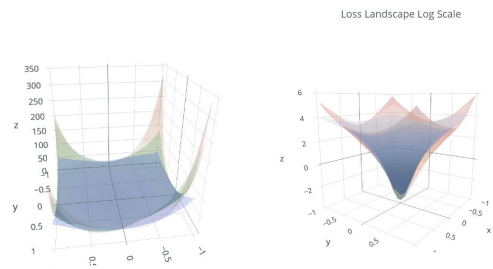
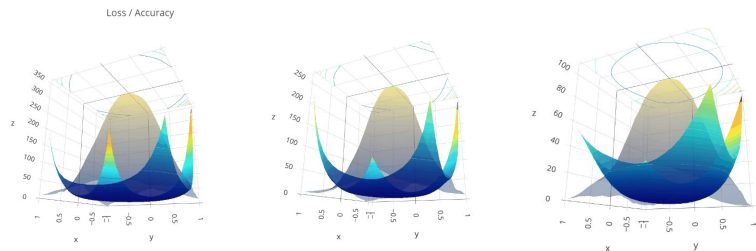


Figure 6: Wide-ResNet-56 on CIFAR-10 both with shortcut connections (top) and without (bottom). The label $k = 2$ means twice as many filters per layer. Test error is reported below each figure.

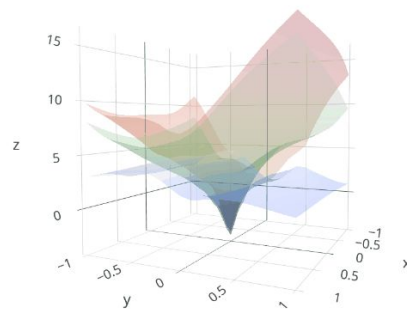
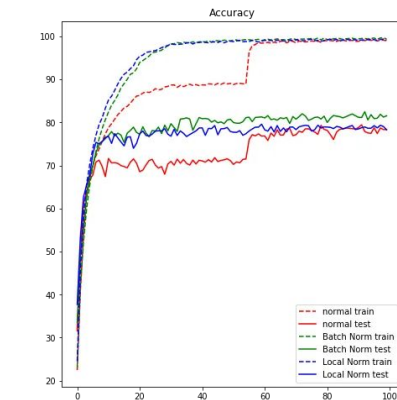
Loss landscape: Caveats

9 layer CNN train on the CIFAR 10 dataset without any normalization, with Batch Normalization and with local response normalization.

We can see that the network with Batch Normalization has achieved the highest performance.



Orthogonal Directions



Principle Directions from Weights

Summary

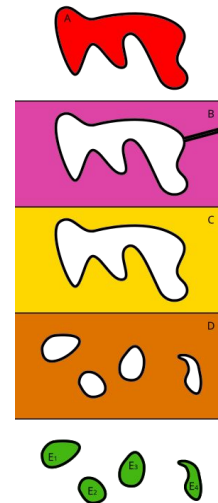
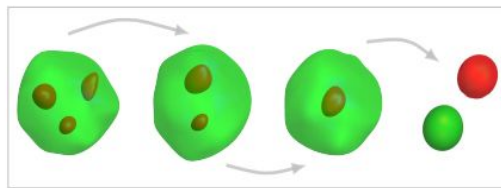
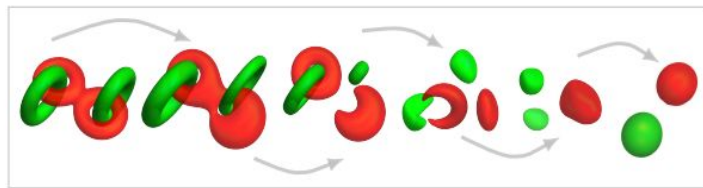
Given enough compute, sample scalars on a fine-grained grid to generate high resolution 3D dimensional views

$$\mathcal{L}(\theta^* + \alpha\delta + \beta\eta)$$

- Smaller mini-batches and weight decay seem to find minimisers θ^* with in wider basins, resulting in less sensitivity to $L(\theta^* + \eta)$ for small η .
- Inclusion of skip connections, i.e. ResNet architectures, and/or increased over parameterisation through greater width appear to give wider less complex landscapes.
- There are theoretical results indicating that width and over parameterisation result in landscapes for which minimisers are simply connected.

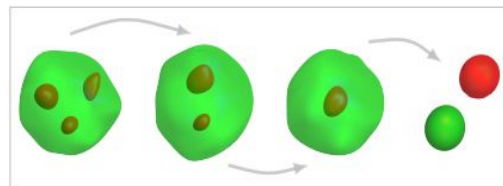
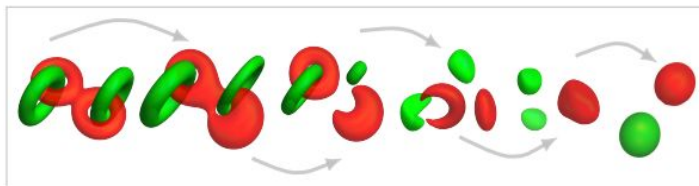
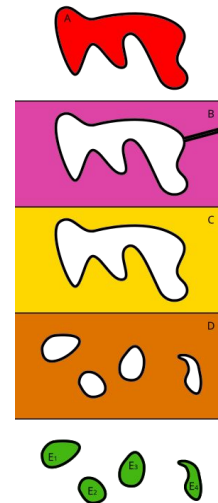
Topology of loss landscape

Topology is the branch of mathematics concerned with the **properties** of a geometric object that are **preserved** under **continuous deformations**, such as stretching, twisting, crumpling, and bending.



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Consider the **Embedding space** of an **IMAGE** as the geometric object of interest, which is modelled through a DNN via **homeomorphic** or **non-homeomorphic** maps based on the task at hand.

Topologically we can study the behaviour/evolution of layer wise embedding space.

For instance, **segmentation** of an image would require a **homeomorphic** map in order to preserve the topology while **classification** requires a **non-homeomorphic** map to collapse the space of images (each class) to a single point/connected component.

Topology of loss landscape

Consider level set of our popular MSE loss function [Result can be extended to other loss fns].

$$\Omega_{\mathcal{L}}(\lambda) = [\theta : \mathcal{L}(\theta; X, Y) \leq \lambda]$$

Of particular interest are the number of connected components, say N_{λ} in $\Omega_{\mathcal{L}}(\lambda)$

If $N_{\lambda} = 1$ for all λ then $\mathcal{L}(\theta; X, Y)$ has no isolated local minima and any descent method can obtain a global minima.

If $N_{\lambda} > 1$ there may be "spurious valleys" in which the minima in the connected component does not achieve the global minima.

Topology of loss landscape

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Of partic

If $N_\lambda =$
method

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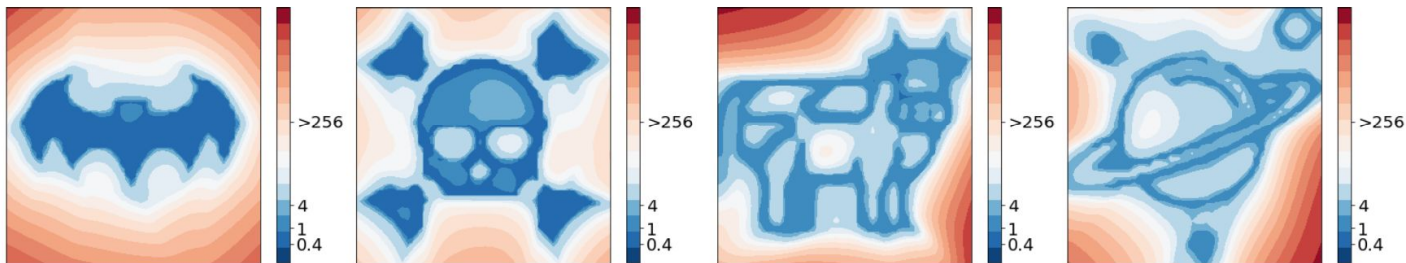
component does not achieve the global minima.

BTW we have PhD openings in the lab if you would like to consider some serious theoretical DL research with lightweight experiments!
Less GPUs more Pen & Paper

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Thanks



(a) Loss surface on FashionMNIST dataset

(b) Loss surface on CIFAR10 dataset

<https://github.com/universome/loss-patterns>